

Research on the Practice of College English Teaching Reform Based on Digital Intelligence Teaching Model

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Abstract

The research focuses on the innovation of college English teaching mode driven by artificial intelligence technology, and uses the New Start College English Comprehensive Course and its supporting AI digital intelligence teaching platform as the practice carrier to build a whole-process intelligent teaching system. Through the five core functional modules of teaching support, learning support, teaching management, evaluation feedback, assessment and monitoring, the platform integrates AI technologies such as language analysis, intelligent scoring, and knowledge graph to achieve accurate push of teaching resources, efficient management of classroom interaction, real-time monitoring of learning behavior, and multi-dimensional intelligent evaluation. Teaching practice shows that through precise planning before teaching, dynamic interaction in teaching, and data-driven optimization after teaching, the AI-empowered closed-loop mechanism of "teaching - learning - managing - evaluating - assessing" has effectively improved the teaching quality and students' language ability, providing a smart teaching paradigm for the digital transformation of higher education.

CCS Concepts

• **General and reference** → Cross-computing tools and techniques; Empirical studies.

Keywords

Artificial Intelligence, College English, Digital Intelligence Teaching Platform

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1 Introduction

In the field of college English teaching, digitally enabled education provides students with a more open, flexible, and personalized learning environment through online courses, virtual laboratories, and intelligent assisted teaching, and promotes the sharing and interactive exchange of teaching resources. The effective integration

of digital technology and university English classroom teaching has promoted the innovation of university English teaching mode [1]. The application of digital technology can effectively integrate online teaching resources, which is more in line with the education and teaching concept of "students as the main body of teaching and teachers as the leader of teaching", which not only realizes the modern teaching classroom, but also improves the teaching effect of the course. The construction of smart teaching models focuses on using advanced technology to improve the efficiency and quality of teaching, while creating a more flexible and personalized learning environment for students [2]. The key to the construction of a smart teaching model is to organically combine technology and teaching, make full use of digital tools and resources, improve the efficiency and quality of education and teaching, and provide teachers and students with a more diverse, interactive, and real-time learning experience, so as to promote the transfer of knowledge and the improvement of learning effectiveness. The theoretical basis of the intelligent teaching model.

2 The theoretical basis of the intelligent teaching model

Constructivism emphasizes that learners actively construct knowledge in their interactions with their environment. The intelligent teaching model provides students with an immersive language learning environment through virtual scene simulation and online collaborative learning, and supports students' independent exploration and knowledge construction [3]. For example, virtual reality technology is used to create English communication scenarios to help students internalize language knowledge in practice. Connectivity considers learning to be the process of connecting nodes to networks. The intelligent teaching platform analyzes students' learning behavior through big data, forms a personalized knowledge network, and promotes information sharing and collaboration among learner [4]. Tools such as MOOCs and learning analysis systems can realize the dynamic connection and personalized push of knowledge. intelligent teaching integrates the advantages of online and offline teaching, realizing the transformation from "teacher-centered" to "student-centered". The intelligent teaching model optimizes the intelligent teaching process and improves the teaching efficiency through AI teaching assistants, intelligent assessment and other technologies [5].

3 Literature Review

The United States, the United Kingdom and other countries have been the first to integrate the concept of smart education into



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foreign language teaching. For example, many colleges and universities in the United States have adopted intelligent language learning platforms Duolingo and Rosetta Stone to adjust learning paths through adaptive learning algorithms to achieve personalized teaching [6]. The Open University of the United Kingdom uses big data to analyze students' learning behavior, optimize curriculum design, and improve teaching and learning effectiveness [7]. In addition, the EU's "Digital Education Action Plan" promotes the application of smart technology in language teaching, emphasizing the use of intelligent translation tools and virtual language partners to improve students' language skills.

The reform of intelligent English teaching in domestic universities shows a trend of diversification. In terms of the application of smart teaching platforms, platforms such as Chaoxing Learning Pass, Rain Classroom, and U Campus are widely used in college English teaching to achieve course resource sharing, real-time interaction, and learning data tracking [8]. In the integration of AI technology: intelligent speech recognition and natural language processing technologies are used in English listening and speaking training, such as iFLYTEK's English listening and speaking intelligent evaluation system, which can correct students' pronunciation errors in real time [9]. In intelligent Teaching Practice, many colleges and universities have explored the model of "online self-directed learning and offline in-depth discussion", using smart classroom tools to carry out flipped classroom teaching and improve student engagement [10].

4 Technical solutions for digital intelligence teaching platform

4.1 Overall architecture and technical characteristics of the platform

With the goal of improving students' abilities, the platform has created an intelligent education system covering the whole process of teaching, learning, managing, evaluating and assessing (see Figure 1). The platform deeply combines advanced technology with English teaching, integrates online and offline resources through intelligent means, and provides comprehensive support for the intelligent English teaching model.

The platform helps teachers accurately adjust their teaching strategies by collecting and analyzing various types of data in the teaching process in real time. The system adopts a modular architecture, and each function can not only operate independently, but also work together to form a complete closed loop of teaching. In terms of technical application, the platform integrates intelligent language processing, speech recognition, knowledge graph and other technologies, adding intelligent tools to traditional classrooms and making teaching more flexible and efficient.

4.2 Core functional module system

4.2.1 Teaching Support Modules. This module provides teachers with a full range of teaching aids, supporting teaching resource management, classroom interaction, and diverse content presentation. Teachers can easily upload courseware, lesson plans, and extension materials, and organize teaching content through mind mapping tools. During the class, the platform provides functions

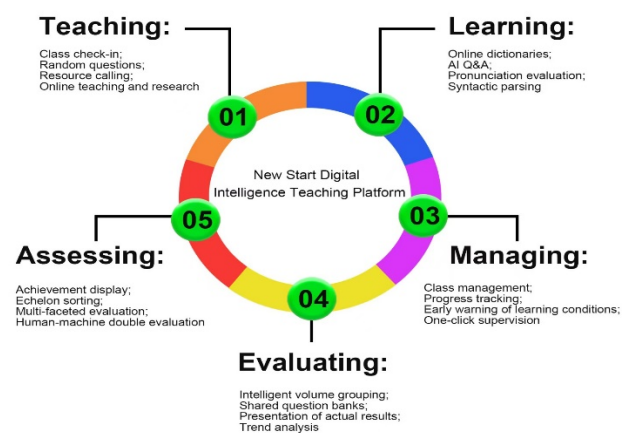


Figure 1: AI digital intelligence teaching platform functions.

such as intelligent check-in, random questioning, and classroom interaction management to help teachers organize teaching efficiently. The teaching content supports audio and video playback, text reading, and visual presentation of knowledge points, making the classroom more vivid and intuitive.

4.2.2 Learning support modules. This module is designed with personalized learning tools designed for students' needs, including basic learning functions and intelligent assistance. Students can use online dictionaries, bookmarks, and other features for self-directed learning at any time. The intelligent assistant can answer English learning questions, analyze sentence structure, and show knowledge point associations through a knowledge graph. The platform also provides functions such as reading training, pronunciation scoring, and dictation practice to help students systematically improve their listening, speaking, reading and writing skills.

4.2.3 Managing Module. This module can realize the digital management of the whole teaching process, including account permission setting, class management and learning process monitoring. Teachers can quickly create classes, manage student groupings, and understand each student's learning status with progress tracking. The system automatically records student interaction data, analyzes learning behavior patterns, provides teachers with a report on the overall learning situation of the class, and supports targeted teaching adjustments.

4.2.4 Evaluation feedback module. This module establishes a diversified evaluation system, combining teacher evaluation and intelligent scoring. Assignment correction supports manual review and automatic correction, and the system can generate personalized comments and learning suggestions. By analyzing the learning data, the platform can generate a profile of the student's ability, showing the personal progress trajectory and the comparison of class learning. At the same time, the platform pays attention to students' language application ability, learning attitude and teamwork performance, and forms a comprehensive evaluation report.

4.2.5 Assessing and monitoring module. The module provides intelligent assessment tools and learning monitoring functions, and

supports a variety of assessment formats such as in-class tests, custom assignments, vocabulary dictation, and oral assessments. The system monitors students' homework completion rate, learning duration and other data in real time, and generates visual charts to help teachers grasp the learning dynamics of the class in a timely manner and optimize the teaching plan.

4.3 Application of intelligent technology in the platform

The platform uses a number of intelligent technologies to improve teaching effectiveness: language parsing technology helps students understand complex sentence structures; The intelligent scoring system provides instant feedback on pronunciation and writing problems; The knowledge graph function connects scattered knowledge points into a system to assist students in building a complete knowledge framework. The learning data analysis module can automatically identify students' weak links and facilitate teachers' supervision; Teachers can use intuitive visual reports to fully grasp the learning dynamics of the class. This series of intelligent applications jointly builds a closed-loop teaching ecology of "data collection, intelligent analysis, and precise guidance", which not only highlights the main position of students, but also gives full play to the enabling role of technology to achieve two-way improvement of teaching and learning.

5 The working mechanism of the intelligent teaching model

5.1 Preparation and planning mechanisms before teaching

Teachers rely on the platform resources and the WeChat virtual teaching and research room to achieve "two-line collaborative lesson preparation", that is, through the virtual teaching and research room every week to carry out cross-school collective lesson preparation, share teaching cases, discuss teaching design, and combine teaching resources such as teaching plan templates, courseware resources, exercise questions and other teaching resources in the platform to quickly generate targeted teaching plans. Teachers can formulate a complete teaching plan covering preview tasks, classroom activities, and after-class extension based on the class learning data, and can also share with teachers to ensure that the pre-class preparation is deeply consistent with the teaching goals.

5.2 Implementation and interaction mechanisms in teaching

Teachers monitor student participation through the platform's check-in, testing, questioning and other functions, and adjust teaching activities in a timely manner. When you find that students are not engaged enough, you can switch to group tasks or interactive activities. In view of the common problems found in the preview, the training resources shared by the teaching and research department can be called for reinforcement. Students can use the platform's 13 student assistance features, including AI conversation, Chinese-English comparison, syntactic analysis, intelligent speech, and more, to receive personalized learning support. The platform can present the data of students' learning status, help teachers grasp

the rhythm of teaching, and achieve a good interaction between teaching and learning.

5.3 Post-teaching evaluation and reflection mechanisms

Teachers can assign a variety of homework assignments, and students can submit them in a variety of formats such as text, audio, and video. The platform's intelligent grading function assists teachers in reviewing assignments and provides timely and professional feedback. At the same time, the AI learning situation analysis system records and mines student learning data, generates learning behavior analysis reports, and helps teachers understand students' learning and knowledge mastery. The virtual department supports teachers to carry out teaching review, share teaching experience, and collectively refine optimization strategies, forming a continuous optimization cycle of "evaluation-feedback-improvement".

6 Effectiveness analysis of curriculum implementation

6.1 Data analysis

Firstly, the analysis of variance function of SPSS20.0 software was used to conduct a one-way analysis of variance between students' college entrance examination scores and level 4 scores under the two teaching modes, so as to explore whether there is a statistically significant difference in students' level 4 examination scores under the two teaching modes. In order to eliminate the differences in the initial English level of students under the two teaching modes, we also analyzed the covariance of the students' grades at level 4 under the two teaching modes using the college entrance examination scores as covariates, as shown in Table 1 and Table 2.

Table 1 presents a comparison of students' CET-4 scores across various components after one-way analysis of variance (ANOVA) and covariance analysis (ANCOVA) under two teaching modes. Table 2 shows the significance P-values of score differences obtained from the ANOVA.

From the data of the one-way ANOVA on teaching modes, it can be observed that the traditional teaching mode outperforms the blended teaching mode by 0.25 points in college entrance examination scores, 3.44 points in total CET-4 scores, 2.09 points in listening, 0.09 points in reading, and 1.26 points in writing. As indicated in Table 2, the P-values for the differences in students' scores across all components under different teaching modes are all less than 0.05, suggesting that there is statistically significant difference in students' CET-4 scores across all components between the two teaching modes.

The ANCOVA with college entrance examination scores as a covariate involves adjusting each student's initial college entrance examination score to the same level before conducting the ANOVA, thereby eliminating the impact of differences in students' English scores in the college entrance examination on the study. The ANCOVA data with college entrance examination scores as the covariate reveals that the differences in students' CET-4 scores across all components between the two modes are further reduced, and the significance P-values become larger. This indicates that after eliminating the influence of college entrance examination scores,

Table 1: Comparison of students' performance data of the two teaching modes (covariance).

Mode	Entrance	Total	Listening	Reading	Writing	Total/ covariance	Listening/ covariance	Reading/ covariance	Writing/ covariance
Traditional standard deviation	114.52 13.36	414.52 57.83	138.40 31.62	144.53 25.53	133.90 16.54	413.784 ^a	138.325 ^a	144.756 ^a	132.686 ^a
Intelligent standard deviation	114.27 12.73	411.08 57.47	136.31 30.20	144.34 25.19	132.64 17.89	411.825 ^a	136.231 ^a	144.823 ^a	132.823 ^a

Table 2: The uni-variate variance (covariance) of the teaching mode analyzes the significance P value of the difference in students' performance.

Mode	Total	Listening	Reading	Writing	Total/ covariance	Listening/ covariance	Reading/ covariance	Writing/ covariance
F value	1.282	2.247	0.326	0.234	0.906	1.902	0.040	0.011
P value	0.003	0.024	0.037	0.029	0.034	0.063	0.024	0.031

Table 3: students' test results on five quantiles.

attributes	Experiment group					Control group				
	10	25	50	75	90	10	25	50	75	90
C6	356	378	427	504	536.3	306.4	343.2	402	467.5	512.6
C6l	102	113	134	173	179.6	91.7	103	129	162	173
C6r	162	167	177	186	205	134	146	159	170	185.6
C6w	88.4	97	117	141	150	84	97	115	132.5	147
C4	432	446	472	531	574	435.4	456	482	518	562
C4l	141	144	156	178	195	140	146.5	159	174	194
C4r	160	166	173	185	202.6	161	171	172	181	194.6
C4w	131	135	145	168	176	131	135	143	162	172

there is still no statistically significant difference in students' CET-4 scores across all components between the two teaching modes; moreover, the significance P-values further decrease, and the differences in scores across all components between the two teaching modes become even larger.

In addition, we conducted a significance test, namely the F-test, on the overall regression equations. It was found that at the significance level of 0.01, the p-values of these four regression models were far less than 0.01. Based on the research results, we hold that the regression equations are significant and the regression relationships are valid. Drawing on the above analysis, we will formulate subsequent linear regression equations to study the correlation between intelligent teaching and English language proficiency.

$$\log(Y_i^\tau) = \alpha + \beta \text{Blend}_i^\tau + \delta \log(\text{Aca}_i^\tau) + \phi \text{Demo}_i^\tau + \varepsilon_i$$

In this formula, $\log$ denotes natural logarithm and τ denotes the quantile point and Y_i^τ stands for the English proficiency of the individual i at the quantile τ in the later stage of the experiment. β preceding the grouping variable Blend_i^τ represents the influence of the intelligent intervention on students i at the quantile τ and is assumed to be positive.

We then used the formula to perform regression analysis for the two sets of scores distributed from the 5th to the 95th percentile. The results are then linearly fitted to obtain a distribution curve representing the effect of intelligent teaching across the entire sample interval, as shown in Figure 3. The results show that the effect of intelligent teaching has a unique pattern, characterized by a greater impact at lower levels of language proficiency, and a downward trend with increased proficiency in the areas of reading, listening, writing, and overall English proficiency. From Table 3, it is not difficult to find that the test scores of the experimental group always exceed those of the control group in five quantiles, which reflects that smart teaching does have advantages over traditional teaching.

By analyzing the performance data of the exam, the learning effect of students' English language skills under the intelligent teaching mode and the traditional teaching mode is explored. The results show that from the overall level, there are significant differences in students' grades at Cet 4 under the two teaching modes, and the effect of intelligent teaching mode is better than that of traditional teaching mode, and there is no decline in teaching quality and learning effect due to intelligent learning. The two teaching

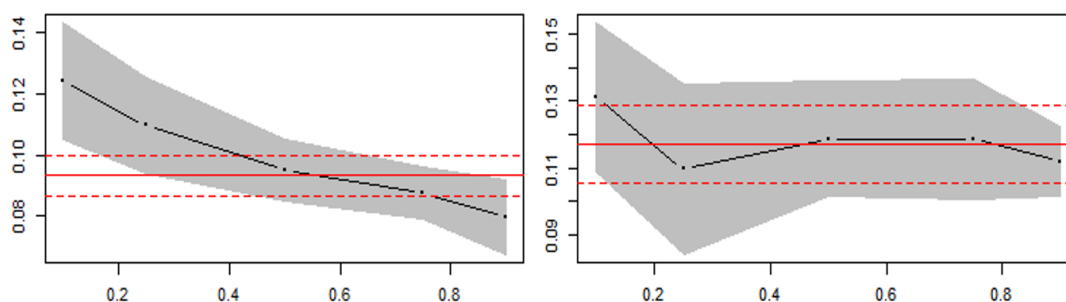


Figure 2: Teaching effect across different language proficiency level groups.

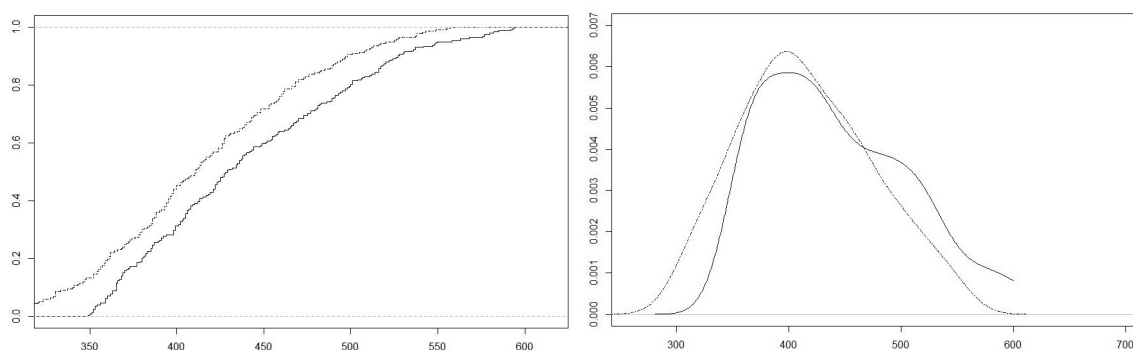


Figure 3: Probability Density Functions and Cumulative Distribution Functions of test scores.

modes have certain differences among students in different proficiency groups. According to Figure 2, the intelligent teaching mode has a greater advantage in the groups with better English proficiency, while the effect of the intelligent teaching mode has a slight advantage in the poor language proficiency groups. The teaching effect of the intelligent teaching mode decreases with the decrease of students' English proficiency.

7 Conclusion

The results show that the implementation of the intelligent teaching mode is of help to improve English teaching efficiency. Its implementation should be determined according to students' English proficiency, learning enthusiasm and learning autonomy. Students with higher English proficiency and high learning autonomy have better learning effects when participating in intelligent teaching, while students with low English proficiency and poor learning autonomy have worse learning effects when participating in smart teaching. It is recommended that teachers should pay attention to cultivating students' learning autonomy while carrying out intelligent teaching, and the higher the students' learning autonomy, the more significant the advantages of intelligent teaching mode. This study discusses the learning effect of smart teaching from the perspective of examination data analysis, and does not cover relevant factors such as teachers' and students' learning initiative.

Considering the differences in college admission policies, the foreign language proficiency of freshmen admitted to higher-level universities is often better than that of ordinary local universities.

Therefore, placing this research result at the macro level of higher education may find that although many domestic universities target the public. The basic English course is or has been tried to teach smartly, but this teaching model is not suitable for all colleges and universities to improve the language of college students. It is a common way to learn skills. For some colleges and universities, because the English of new students has reached a high level, the focus is on grammar and The smart teaching of college English may no longer be effective, and the practice of smart teaching at this time should jump out of the scope of general English. The combination of in-depth academic knowledge and professional background, and more thinking from the level of textbooks and curriculum design, can be guaranteed. These students have greater language learning opportunities and room for growth.

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